

07 - PCB Layout Rules - Rev B2

Power-input region

Rule	Requirement
Q1 placement	Place AO3401A immediately after J1 5V_IN. Route pin 3 to input, pin 2 to the protected rail, and pin 1 to RG.
Gate pull-down	Place RG = 100 kΩ close to Q1 gate and return it directly to GND_HOST.
FB1 placement	Place FB1 after Q1 and before the DC/DC input capacitors. Keep the noisy side and filtered side copper distinct.
No PPTC / TVS	Do not populate footprints for an on-board PPTC or input TVS in the Rev B2 production layout. The design assumes
DC/DC loop	Place 4.7 μF and 100 nF directly at PS1 input pins. Place 10 μF and 100 nF directly at isolated output pins.
Analog separation	Keep PS1 and FB1 switching/noise region away from SUM1, SUM2, R1B, R24/R33/R50, and the photodiode pin
Ground domains	Use separate copper for GND_HOST and GND_ISO; no stitching or test jumper between them.
Selector network	Place SW1 and feedback resistors close to U2. Make the three switch positions clearly legible on silkscreen.
Creepage	Maintain barrier creepage/clearance based on the intended working voltage, pollution degree, and enclosure.
Cleanliness	Clean flux around high-impedance photodiode nodes and precision feedback components.

Release checklist

Verify Q1 source/drain orientation against the exact manufacturer datasheet; confirm FB1 current rating; confirm there is no copper connection across the isolation barrier; and visually verify that the PCB contains no PPTC or input-TVS population in the Rev B2 assembly variant.